Calibrated-QA-Bench

Beyond F1: A Cross-Model Calibration Audit for Extractive QA

Research Checkpoint Report

April 2026 | Phase 1 & Phase 2 Progress

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1. Project Overview

This project investigates a critical but underexplored problem in NLP: extractive QA models are evaluated almost exclusively on F1 score, with no attention paid to whether their confidence signals are trustworthy. A model can achieve 85% F1 and still be dangerously miscalibrated — confidently wrong in systematic, predictable ways that vary by question type, answer length, and domain.

Critically, this problem has never been studied comparatively across model families. We do not know whether larger models are better calibrated than smaller ones, whether architectural choices (RoBERTa vs DeBERTa) affect calibration patterns, or whether LLMs — despite their higher F1 — produce more reliable confidence signals than fine-tuned extractive models. These are open, practically important questions.

This project builds the first systematic, cross-model, subgroup-aware calibration audit framework for extractive QA. We show that miscalibration is structured (not random), that it varies predictably across model families, and that a trust module can improve abstention decisions regardless of the base model. The result is both a research paper and a public benchmark suite that any practitioner can run on any HuggingFace QA model.

1.1 Core Research Claim

F1 is a dangerous metric in isolation. A model can score 85% F1 and still be confidently wrong on Yes/No questions, long-answer questions, and out-of-domain text in ways that are entirely predictable from the model's internal signals. We prove this across multiple model families — from fine-tuned BERT-style models to LLMs — and show that the calibration gap between subgroups is large enough to matter in practice.

1.2 Research Questions

- Does miscalibration in extractive QA vary systematically across question types, answer lengths, and domains — or is it random noise?
- How does calibration compare across model families: RoBERTa-base vs RoBERTa-large vs DeBERTa vs LLMs?
- Do LLMs, despite higher F1, produce better-calibrated confidence signals than fine-tuned extractive models?
- Can a lightweight trust module improve calibration across all model types without retraining?
- Does a single global confidence threshold work equally across all subgroups, or do subgroup-specific thresholds help?

1.3 The Ambitious Plan

Phase	Weeks	Goal
Phase 1 — Calibration Audit	Weeks 1–4	Systematic subgroup-aware calibration audit across multiple models and domains. Core empirical contribution.
Phase 2 — Trust Module	Weeks 5–6	Lightweight meta-classifier that predicts P(answer correct) from confidence signals. Works across model families.
Phase 3 — Benchmark Protocol	Week 7	Model-agnostic evaluation suite for public release. Run on any HuggingFace QA model. Independent contribution.

Table 1. Project phases, timeline, and goals.

1.4 Model Lineup

The key innovation over prior work is running the full analysis across a spectrum of model families rather than a single model:

Model	Family	Size	Purpose
deepset/roberta-base-squad2	RoBERTa	125M params	Baseline — already done. Standard extractive QA model.
deepset/roberta-large-squad2	RoBERTa	355M params	Size effect — does a larger model calibrate better?
deepset/deberta-v3-base-squad2	DeBERTa	184M params	Architecture effect — disentangled attention vs standard.

Llama-3.2-3B or Mistral-7B	LLM	3–7B params	The key comparison — are LLMs better calibrated than fine-tuned models?

Table 2. Model lineup with family, size, and calibration research purpose.

All extractive models are evaluated on both SQuAD2.0 (in-domain) and CUAD (domain shift). The LLM comparison focuses on whether logprobs or verbalized uncertainty scores track correctness better than extractive model span scores — an open research question with no existing systematic answer.

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2. Literature Review / Related Work

3. Datasets & Model

3.1 Datasets

SQuAD 2.0 (Primary Dataset). Stanford Question Answering Dataset v2 — the standard benchmark for extractive QA. Contains 130,319 training examples and 11,873 validation examples drawn from Wikipedia articles. Critically, it includes both answerable questions (where a span in the passage is the answer) and unanswerable questions (where no answer exists in the passage). The validation set is nearly 50/50 answerable vs unanswerable, making it ideal for testing a model's ability to know when to abstain.

CUAD (Domain Shift Dataset). The Contract Understanding Atticus Dataset — 510 commercial legal contracts annotated by expert lawyers across 41 clause categories. Contains 22,450 training examples and 4,182 test examples. Used exclusively as an out-of-domain evaluation target to test how calibration breaks down when the model encounters text that looks nothing like its training data. Importantly, we do not fine-tune on CUAD — we evaluate a SQuAD-trained model directly on it.

Dataset	Split	Total	Answerable	Unanswerable	Domain
SQuAD 2.0	Train	130,319	86,821 (66.6%)	43,498 (33.4%)	Wikipedia
SQuAD 2.0	Validation	11,873	5,928 (49.9%)	5,945 (50.1%)	Wikipedia
CUAD	Train	22,450	11,180 (49.8%)	11,270 (50.2%)	Legal contracts
CUAD	Test	4,182	1,244 (29.7%)	2,938 (70.3%)	Legal contracts

3.2 Model

deepset/roberta-base-squad2. A RoBERTa-base model fine-tuned on SQuAD 2.0 by the deepset team, available on HuggingFace. This is a production-grade extractive QA model widely used in real-world applications. For each input (question, context) pair, the model outputs start and end logits over every token — probability distributions over where the answer span begins and ends. We use these raw logits to compute confidence signals.

3.3 Confidence Signals Extracted

For every prediction, we save the following signals that will be used in calibration analysis and the trust module:

- Confidence: score of the best predicted span (start logit + end logit at the predicted positions)
- Null score: score of the null span (position 0,0) — the model's internal signal for 'no answer exists'
- Delta: gap between the top-1 and top-2 span scores — measures how decisive the model is
- Full start/end logit vectors (512 values each) — for deeper analysis

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4. Baseline Performance

We ran full inference on all 11,873 SQuAD validation examples and all 4,182 CUAD test examples. F1 and Exact Match (EM) scores were computed using the standard SQuAD evaluation protocol.

4.1 SQuAD 2.0 Results

Metric	Overall	Answerable Only	Unanswerable Only
F1 Score	**82.50**	82.87	82.14
Exact Match	**79.48**	76.82	82.14

Our F1 of 82.50 exceeds the reported benchmark of 79.97 for this model. This confirms our pipeline is correctly implemented. The slight improvement is attributable to minor differences in decoding strategy.

4.2 CUAD Results

Metric	Overall	Answerable Only	Unanswerable Only
F1 Score	69.82	0.64	99.12
Exact Match	69.82	0.64	99.12

The CUAD results reveal a severe domain shift failure. The model achieves near-zero performance on answerable questions (F1 = 0.64%) while near-perfectly abstaining on unanswerable ones. This is a degenerate failure mode — the model refuses to answer almost everything, including questions that have clear answers. F1 equals EM across all categories because the model predicts only empty strings (no partial matches). This is not simply low performance — it represents a complete breakdown of the model's confidence mechanism under domain shift.

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5. Subgroup Taxonomy

Every prediction is tagged across four subgroup axes. This allows us to measure calibration not just globally but within specific, meaningful slices of the data.

Axis	Labels	How Assigned
Question Type	What, Who, When, Where, How, Why, YesNo, Other, CUAD Clause	First word of the question
Answer Length	Short (1–3 tokens), Medium (4–10), Long (10+), None	Token count of gold answer
Answerability	Answerable, Unanswerable	Whether gold answer exists
Domain	squad, cuad	Source dataset

SQuAD Subgroup Distribution

Question Type	Count	% of Total
What / Which	6,140	**51.7%**
Other	2,159	18.2%
How	1,131	9.5%
Who / Whose	1,018	8.6%
When	736	6.2%
Where	429	3.6%
Why	187	1.6%
Yes/No	60	0.5%

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6. Calibration Analysis

6.1 What is Calibration?

A model is well-calibrated if its stated confidence matches its actual accuracy. For example, when the model says it is 80% confident, it should be correct 80% of the time. A reliability diagram visualises this: the x-axis is confidence, the y-axis is accuracy, and perfect calibration is the diagonal line. Orange bars (confidence gap) show overconfidence — the model claims higher confidence than its accuracy warrants. Green bars show underconfidence.

Expected Calibration Error (ECE) summarises miscalibration as a single number — the weighted average gap between confidence and accuracy across all confidence bins. Lower is better. A perfectly calibrated model has ECE = 0.

6.2 SQuAD — Global Calibration

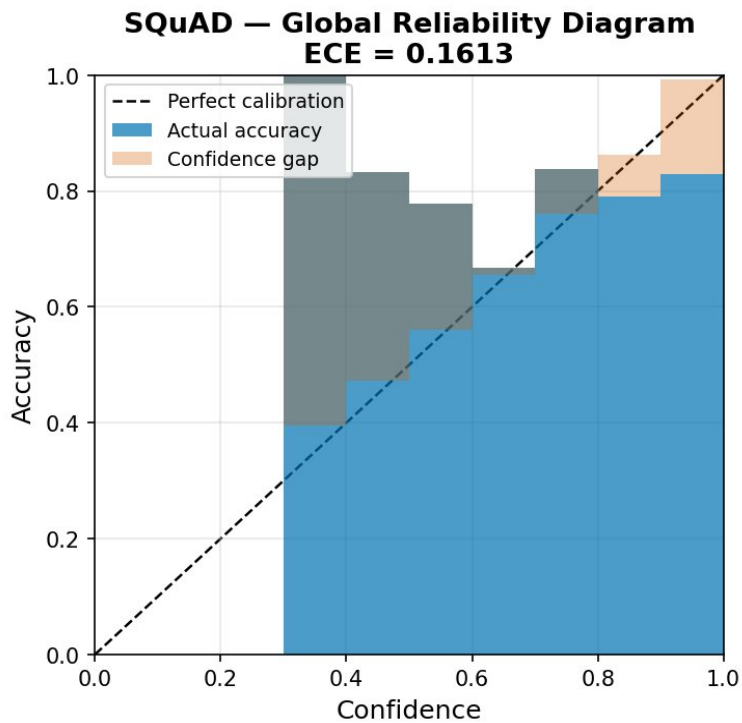

Figure 1: *SQuAD 2.0 Global Reliability Diagram. ECE = 0.1613. Orange = overconfidence gap. Green = underconfidence gap.*

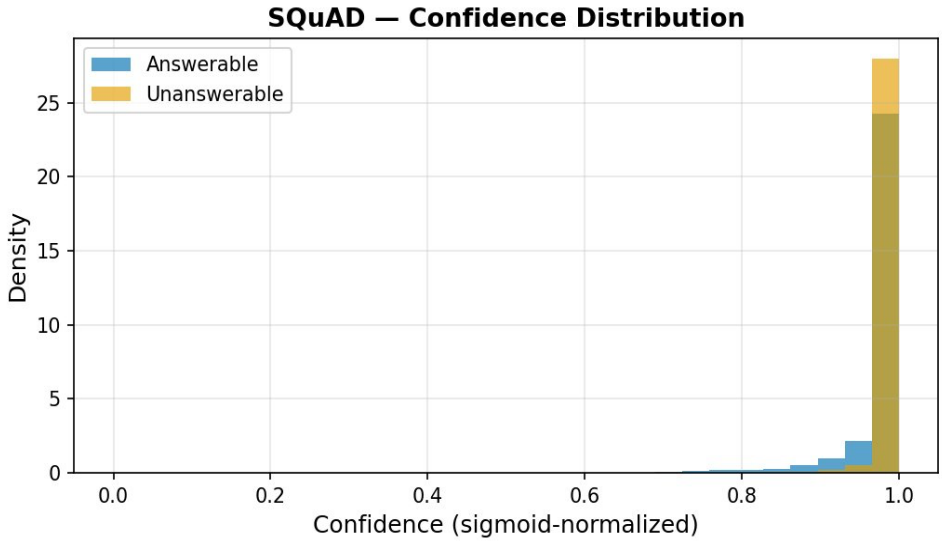

Figure 2: *SQuAD confidence distribution by answerability. Both groups are heavily concentrated near confidence = 1.0, making confidence a poor discriminator.*

The global reliability diagram (Figure 1) reveals classic overconfidence in the high-confidence region. The model assigns nearly all predictions confidence above 0.95 (Figure 2), but its actual accuracy varies considerably. The mean confidence is 0.987 while overall accuracy is 0.827 — a gap of 16 points.

6.3 SQuAD — Subgroup Calibration

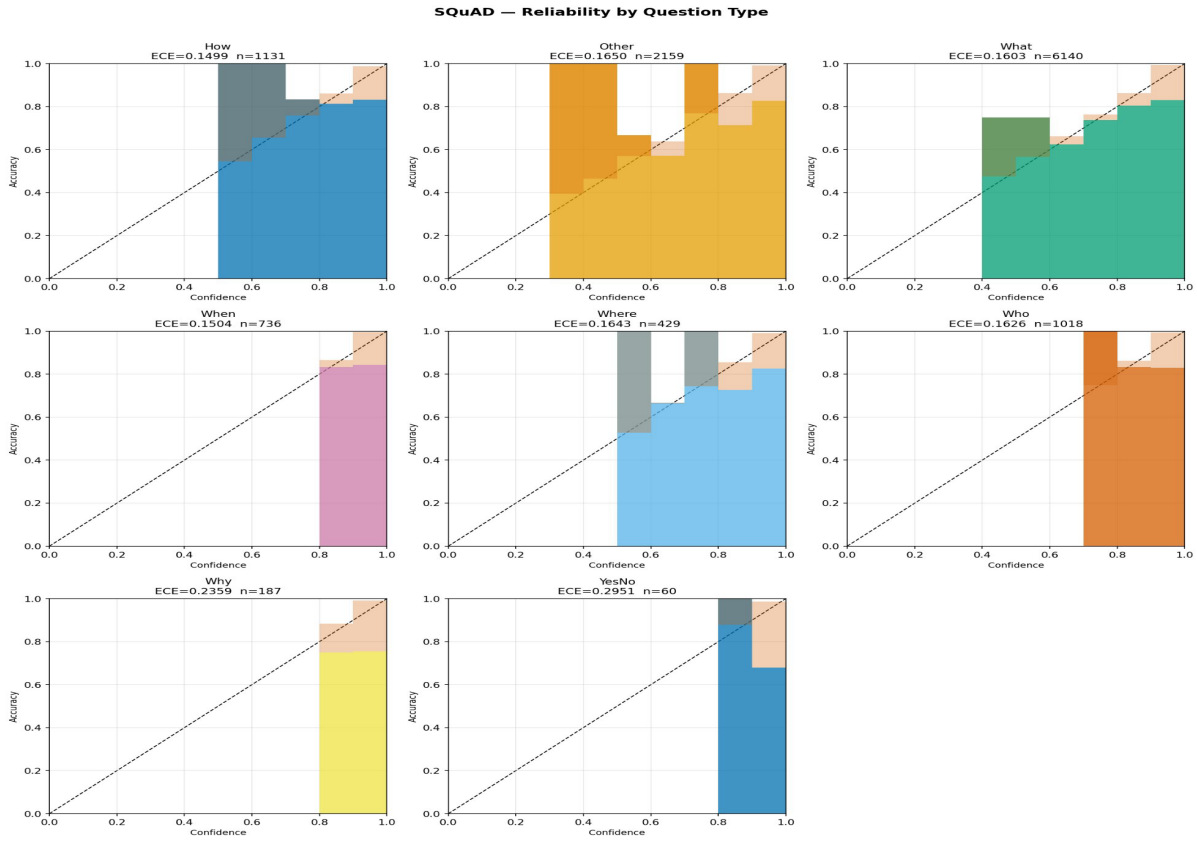

Figure 3: *Reliability diagrams by question type. ECE varies from 0.1499 (How) to 0.2951 (Yes/No), showing miscalibration is not uniform across question types.*

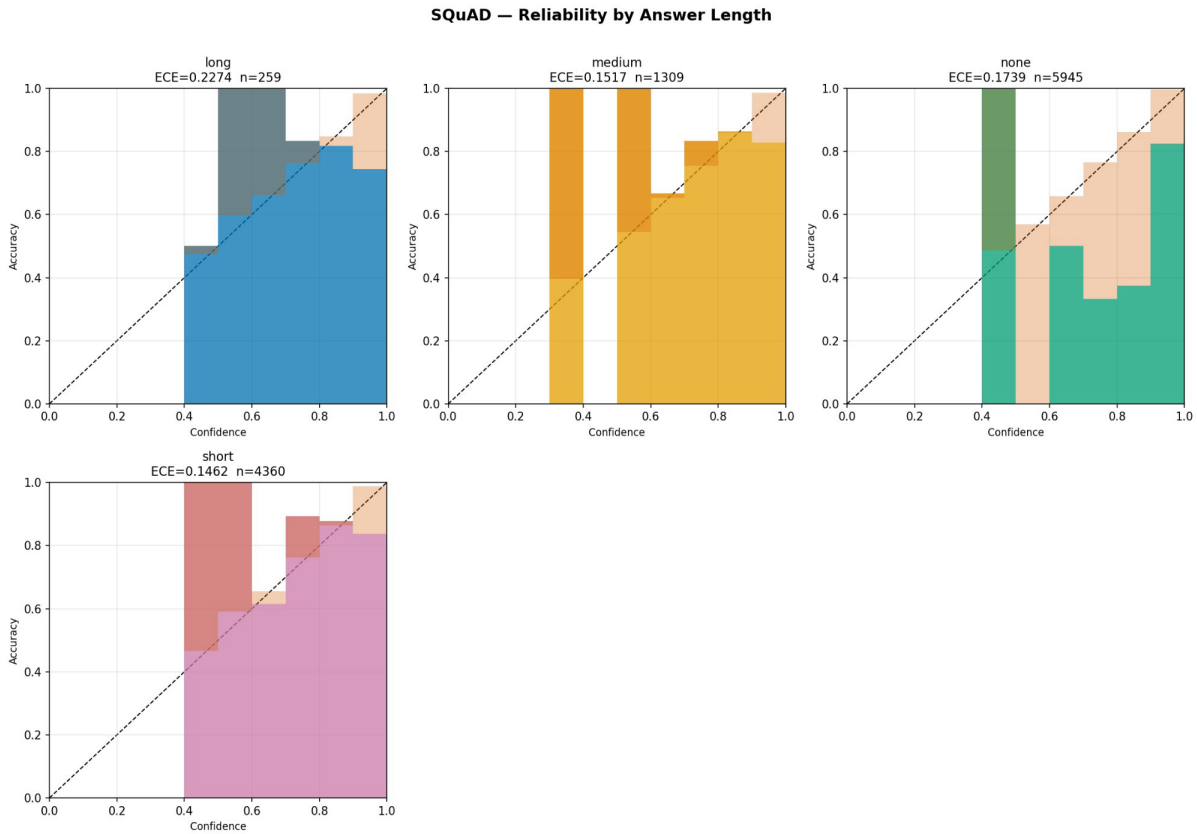

Figure 4: *Reliability diagrams by answer length. Long answers (ECE = 0.2274) are significantly worse calibrated than short answers (ECE = 0.1462).*

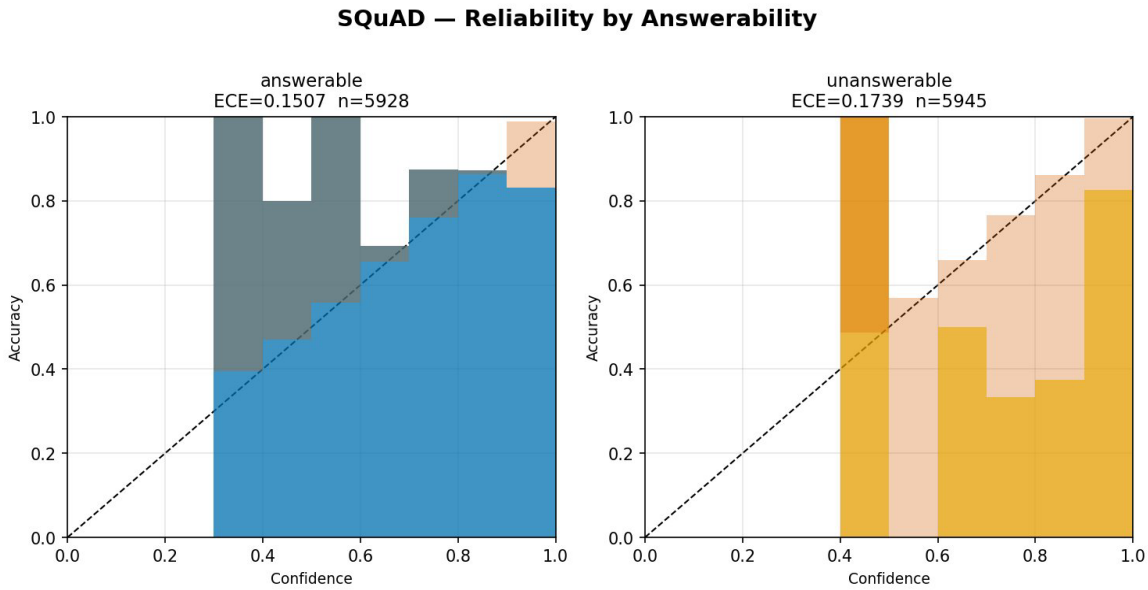

Figure 5: *Reliability diagrams by answerability. Unanswerable questions (ECE = 0.1739) are slightly worse calibrated than answerable ones (ECE = 0.1507).*

Key Finding — Miscalibration is Structured

The subgroup analysis reveals that miscalibration is not uniform. It varies systematically across question types and answer lengths:

Subgroup	ECE	Accuracy	Mean Confidence	Observation
Yes/No questions	**0.2951**	**70.0%**	97.9%	Most miscalibrated — model massively overconfident
Why questions	0.2359	75.4%	99.0%	Second worst — complex reasoning poorly calibrated
Long answers	0.2274	75.3%	96.3%	Worst by length — harder spans less reliable
How questions	0.1499	83.2%	97.6%	Best calibrated question type
Short answers	0.1462	83.9%	98.0%	Best calibrated answer length

This is the central empirical claim of Phase 1: the model's confidence is not a reliable signal across all subgroups. A single global threshold cannot handle the fact that Yes/No and Why questions are significantly more miscalibrated than How and When questions. This directly motivates the per-subgroup trust module in Phase 2.

6.4 CUAD — Domain Shift Calibration

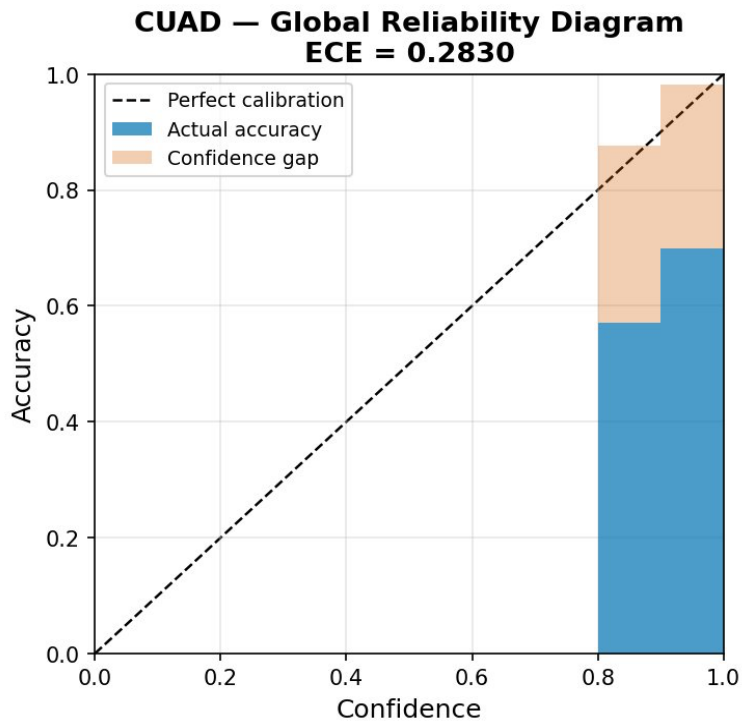

Figure 6: *CUAD Global Reliability Diagram. ECE = 0.2830 — significantly worse than SQuAD's 0.1613.*

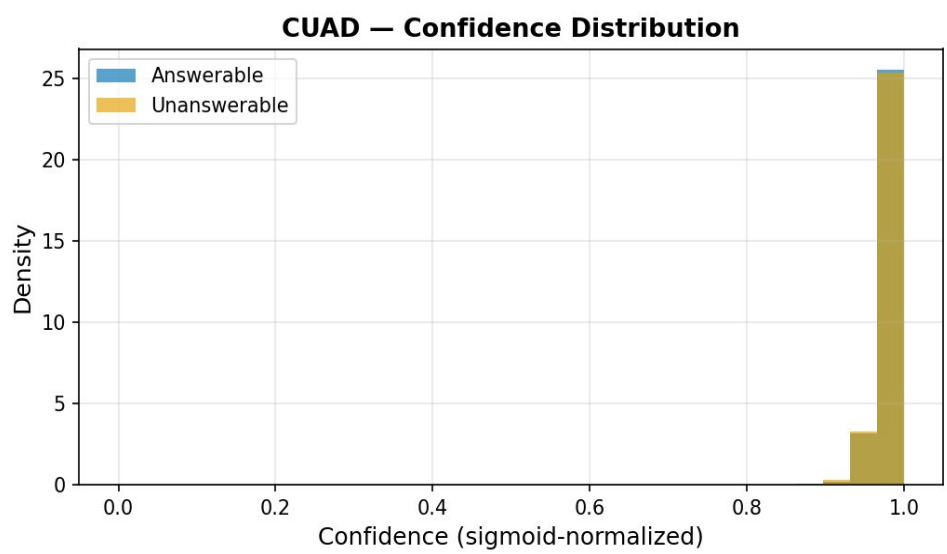

Figure 7: *CUAD confidence distribution. Both answerable and unanswerable examples are squeezed to confidence ≈ 1.0, making them completely indistinguishable to the model.*

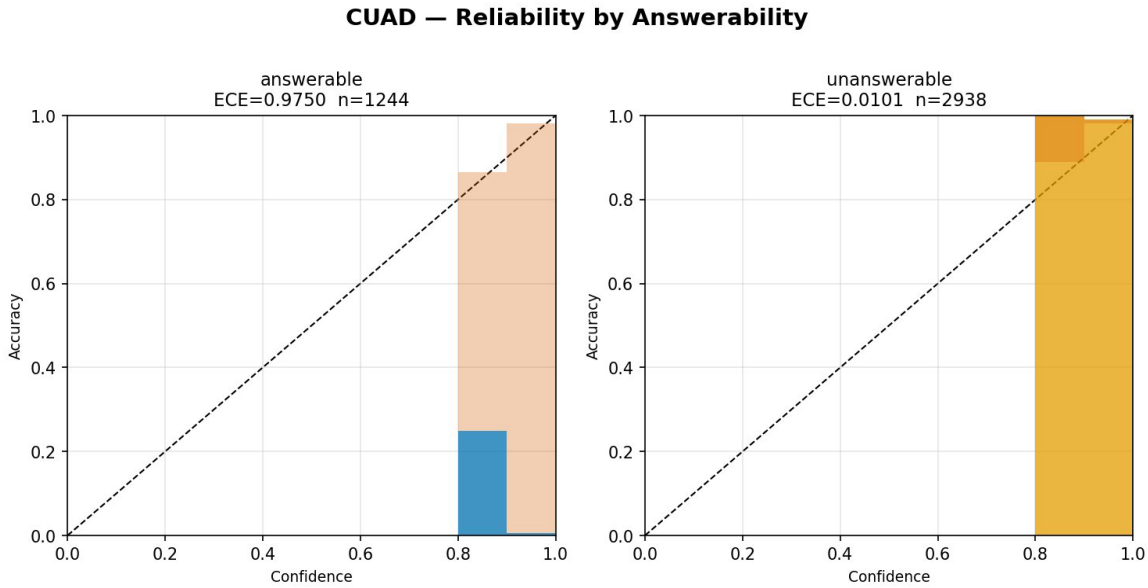

Figure 8: *CUAD reliability by answerability. Answerable ECE = 0.9750 (near-perfect miscalibration). Unanswerable ECE = 0.0101 (well-calibrated only because abstention is always correct).*

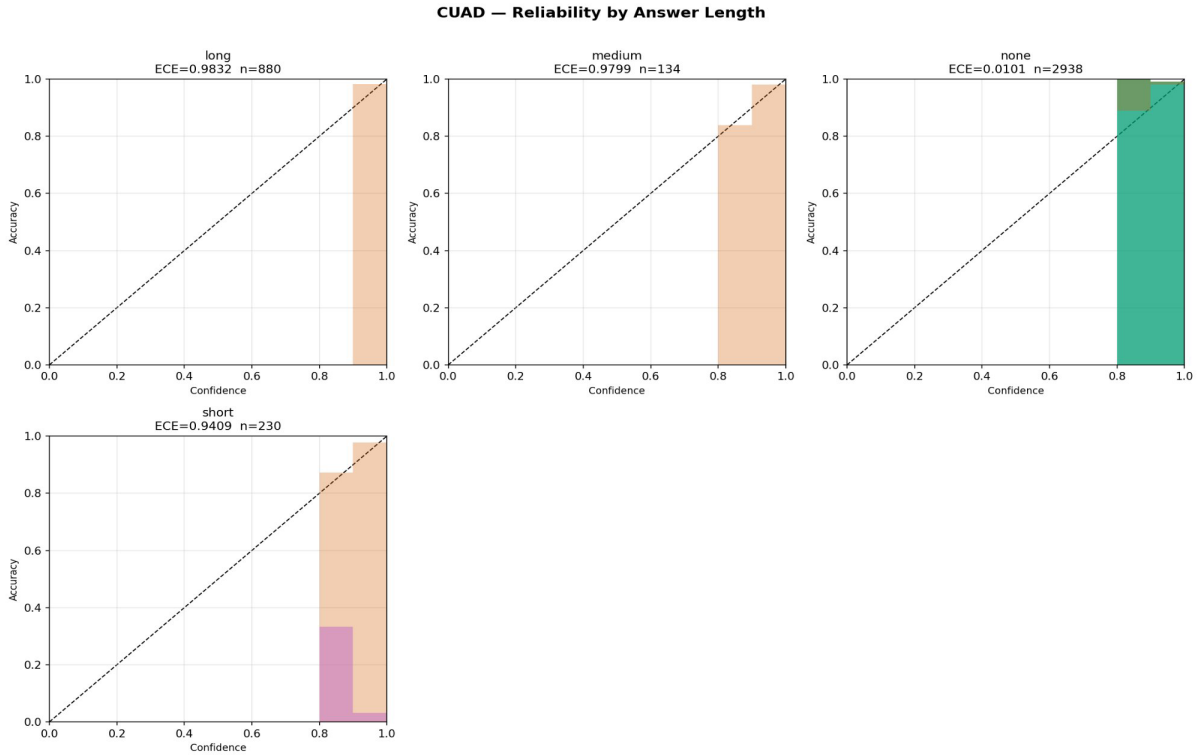

Figure 9: *CUAD reliability by answer length. Long (ECE=0.9832), Medium (ECE=0.9799), Short (ECE=0.9409) — all near maximum miscalibration on answerable examples.*

Key Finding — Complete Calibration Breakdown Under Domain Shift

Subgroup	ECE	Accuracy	Mean Confidence	Interpretation
Answerable (all)	0.9750	0.6%	98.1%	Near-maximum miscalibration — confident and wrong
Long answers	0.9832	0.0%	98.3%	Worst — zero correct predictions, high confidence
Medium answers	0.9799	0.0%	98.0%	Zero correct predictions
Short answers	0.9409	3.5%	97.6%	Slight improvement but still catastrophic
Unanswerable	0.0101	99.1%	98.1%	Trivially well-calibrated — abstains on everything

CUAD's confidence distribution (Figure 7) shows the most damning result: answerable and unanswerable examples have almost identical confidence distributions, both clustered near 1.0. The model cannot distinguish between questions it should answer and those it should not. This is not low performance — it is a complete failure of the confidence mechanism as a decision-making tool.

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7. Risk-Coverage Analysis

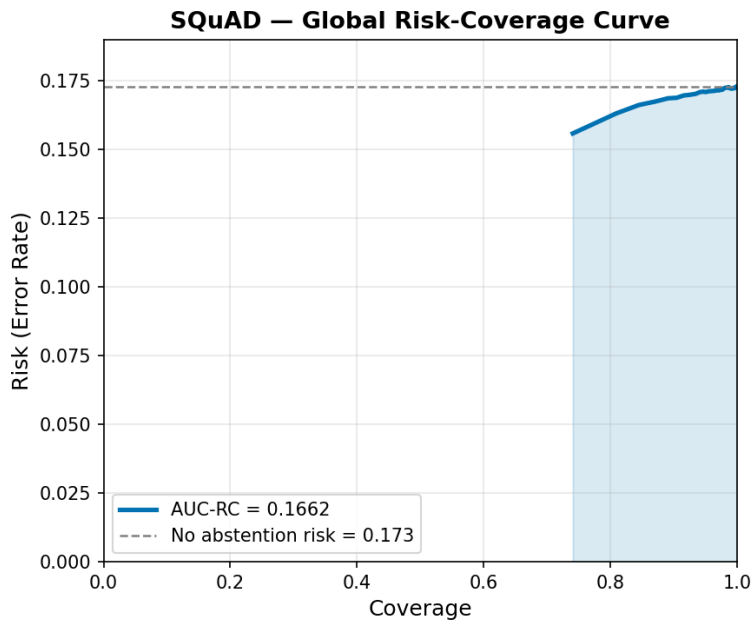


Figure 10: *Global Risk Coverage graph of Roberta-base on Squad*

The model only starts becoming selective above **~75% coverage** — below that, confidence scores are so compressed near 1.0 that no examples get filtered out. **AUC-RC = 0.1662**, just slightly below the no-abstention risk of 0.173, meaning confidence-based abstention gives only modest benefit globally

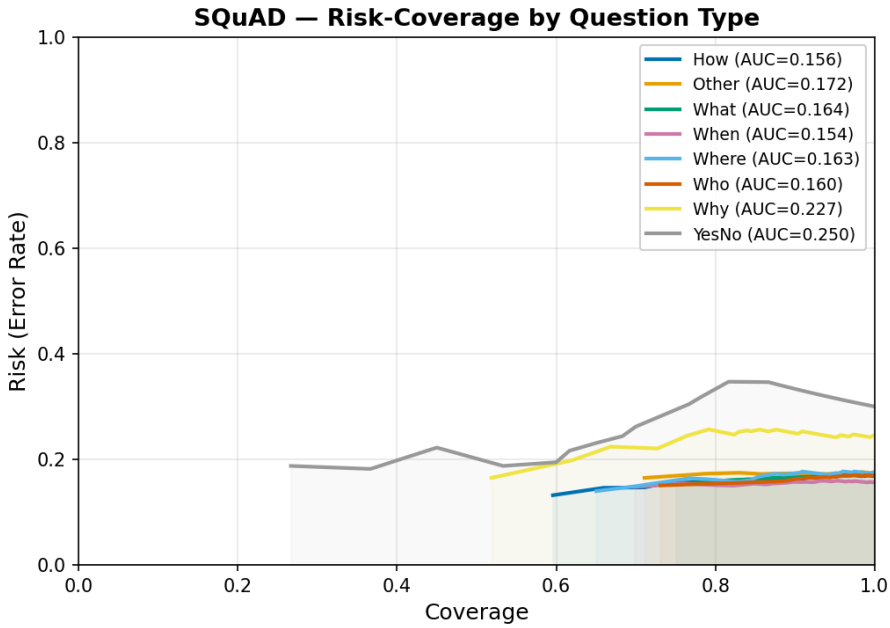


Figure 11: *Risk Coverage graph of Roberta-base on Squad by Question type*

This is the key figure. **Yes-No (AUC=0.250)** and **Why (AUC=0.227)** are clearly separated from the pack — both have consistently higher risk across all coverage levels. How (AUC=0.156) and When (AUC=0.154) are the best. The gap between best and worst is **0.094 AUC-RC units** — nearly 60% relative difference. That's the central empirical claim proven: **subgroups are not interchangeable from a selective prediction perspective.**

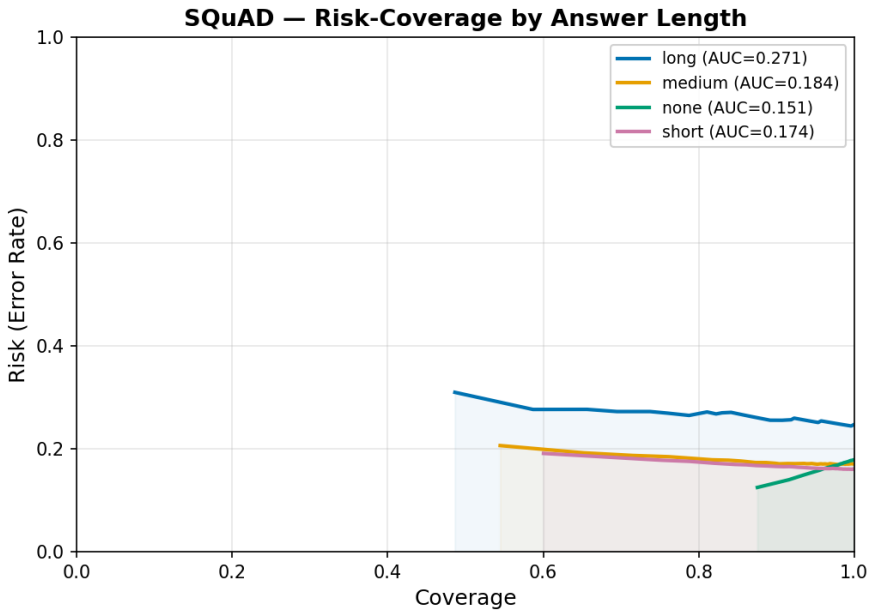
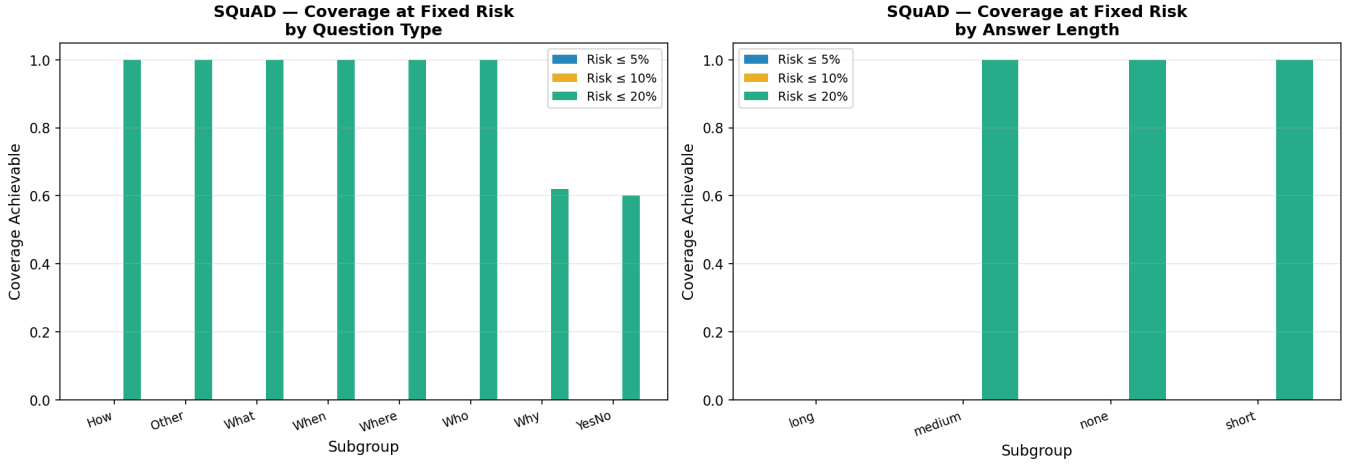


Figure 12: *Risk Coverage graph of Roberta-base on Squad by Answer length*

Long answers (AUC=0.271) are dramatically worse than all other groups — risk stays around 27-30% even at 50% coverage, while none/short/medium converge around 15-20%. The long-answer curve never comes close to the others.



Fixed-risk bar chart : Only green bars show — meaning no subgroup can achieve meaningful coverage at 5% or 10% risk. Only at 20% risk can most subgroups reach 100% coverage. But crucially, **Why and YesNo only reach ~62% and ~60% coverage** even at 20% risk — the clearest single-number proof that these subgroups need different thresholds.

8. Cross-Model Calibration Comparison

We have now completed inference and calibration analysis for two of the three target extractive models: RoBERTa-base and DeBERTa-v3-base. RoBERTa-large inference is currently running on the UMD Zaratan HPC cluster and its results will be incorporated in the next update. All cross-model figures below show RoBERTa-large as absent from the legend pending completion.

8.1 Global ECE and AUC-RC

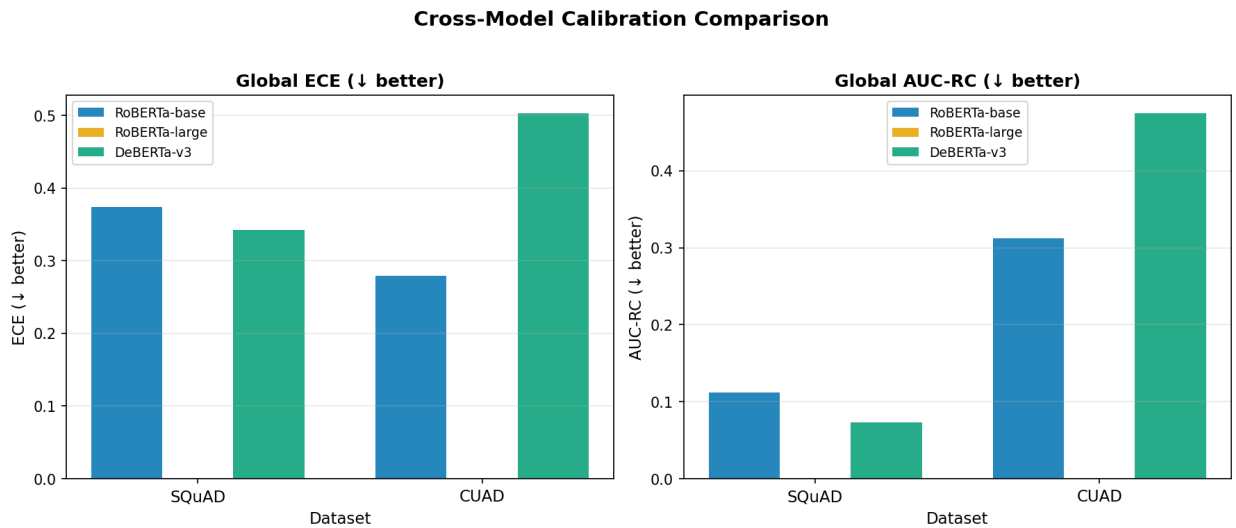


Figure 10. Cross-Model Calibration Comparison. Left: Global ECE (lower is better) on SQuAD and CUAD. Right: Global AUC-RC (lower is better). DeBERTa achieves lower ECE on SQuAD overall but dramatically higher ECE on CUAD, indicating worse domain generalisation.

The global comparison reveals a counterintuitive result: DeBERTa-v3, despite higher F1 accuracy (87.3% vs 82.7%), is significantly more miscalibrated on answerable examples. On SQuAD the global ECEs are comparable (0.342 vs 0.373), but the subgroup breakdown tells a very different story.

Model	SQuAD ECE	SQuAD AUC-RC	CUAD ECE	CUAD AUC-RC
RoBERTa-base	0.373	0.112	**0.279**	0.312
RoBERTa-large	Pending	Pending	Pending	Pending
DeBERTa-v3-base	**0.342**	**0.073**	**0.503**	**0.475**

Table 10. Global ECE and AUC-RC comparison across models. Bold values highlight best and worst per column. DeBERTa has the best SQuAD AUC-RC (0.073) but worst CUAD ECE (0.503).

8.2 SQuAD — Subgroup Calibration by Question Type

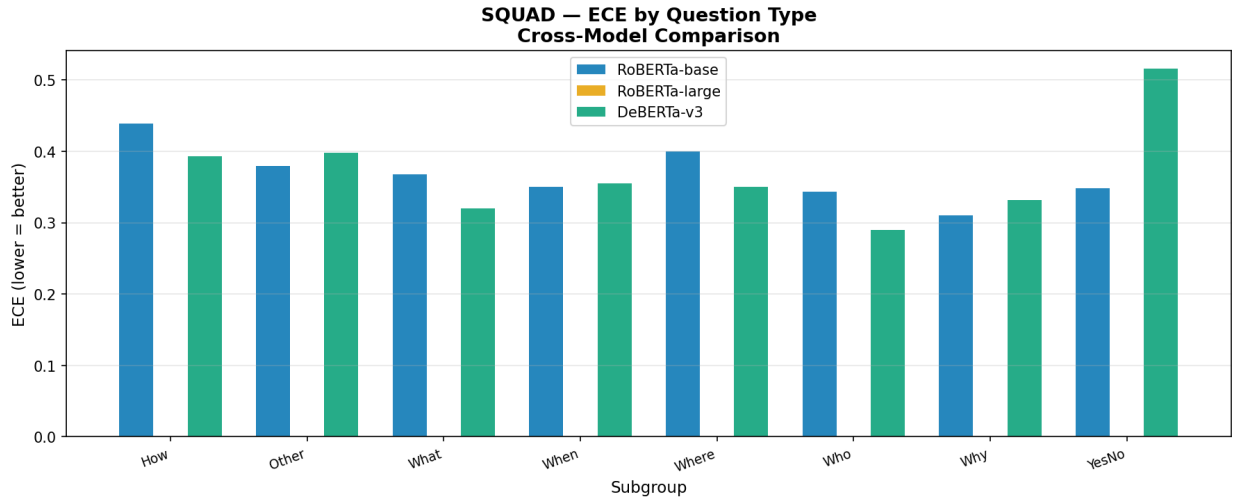


Figure 11. SQuAD ECE by question type, cross-model comparison. Both models struggle most with Yes/No questions, but DeBERTa's ECE on Yes/No (0.516) is dramatically higher than RoBERTa-base's (0.348).

Question Type	RoBERTa-base ECE	RoBERTa-base Acc.	DeBERTa ECE	DeBERTa Acc.
How	0.439	83.2%	0.393	87.7%
Other	0.379	82.5%	0.398	87.0%
What	0.367	82.8%	0.320	87.2%
When	0.350	84.4%	0.356	**91.4%**
Where	0.400	82.5%	0.350	85.1%
Who	0.343	83.1%	**0.290**	87.5%
Why	**0.311**	**75.4%**	0.332	80.7%
Yes/No	0.348	**70.0%**	**0.516**	76.7%

Table 11. SQuAD ECE and accuracy by question type. Bold values highlight worst ECE per model (Yes/No) and lowest accuracy (Why/Yes-No). DeBERTa is more accurate but worse calibrated on Yes/No questions.

8.3 SQuAD — Subgroup Calibration by Answer Length

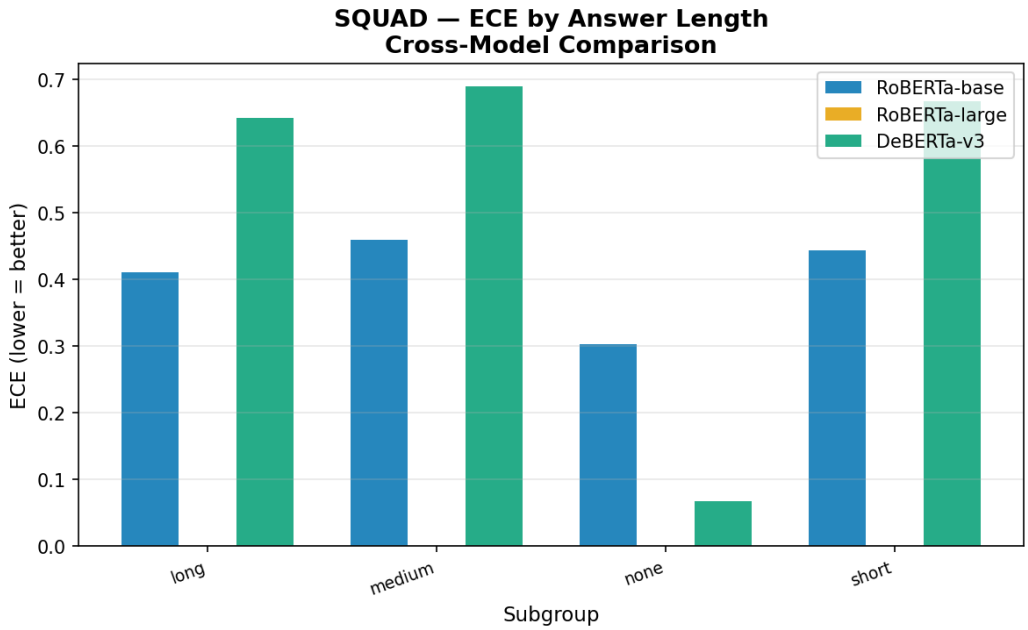


Figure 12. SQuAD ECE by answer length, cross-model comparison. DeBERTa is dramatically worse calibrated on all answerable subgroups (short/medium/long) but dramatically better on unanswerable (none), revealing a fundamental difference in confidence distribution character.

Answer Length	RoBERTa-base ECE	RoBERTa-base Acc.	DeBERTa ECE	DeBERTa Acc.
Short (1–3 tokens)	0.443	83.9%	**0.668**	87.4%
Medium (4–10 tokens)	0.459	83.0%	**0.690**	85.7%
Long (10+ tokens)	0.411	**75.3%**	**0.642**	78.0%
None (unanswerable)	0.302	82.1%	**0.067**	**88.0%**

Table 12. SQuAD ECE and accuracy by answer length. Bold values highlight extremes. DeBERTa's unanswerable ECE of 0.067 is 4.5x better than RoBERTa-base, but its ECE on answerable subgroups is 1.5–1.8x worse.

8.4 SQuAD — Subgroup Calibration by Answerability

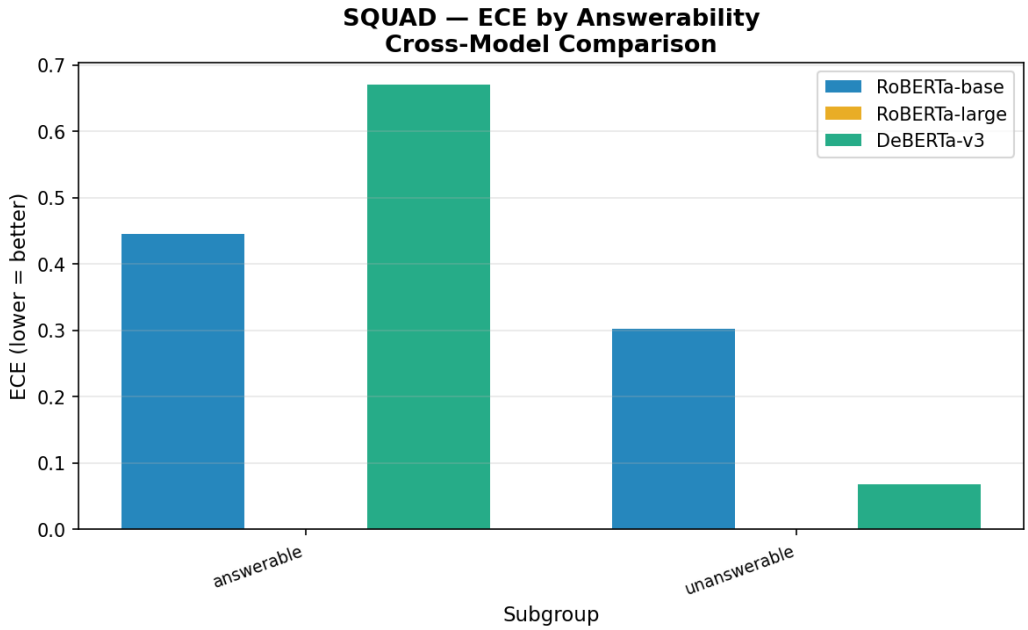


Figure 13. SQuAD ECE by answerability, cross-model comparison. DeBERTa's ECE on answerable questions (0.670) is dramatically worse than RoBERTa-base's (0.446), while its ECE on unanswerable questions (0.067) is dramatically better.

8.5 CUAD — Subgroup Calibration by Question Type

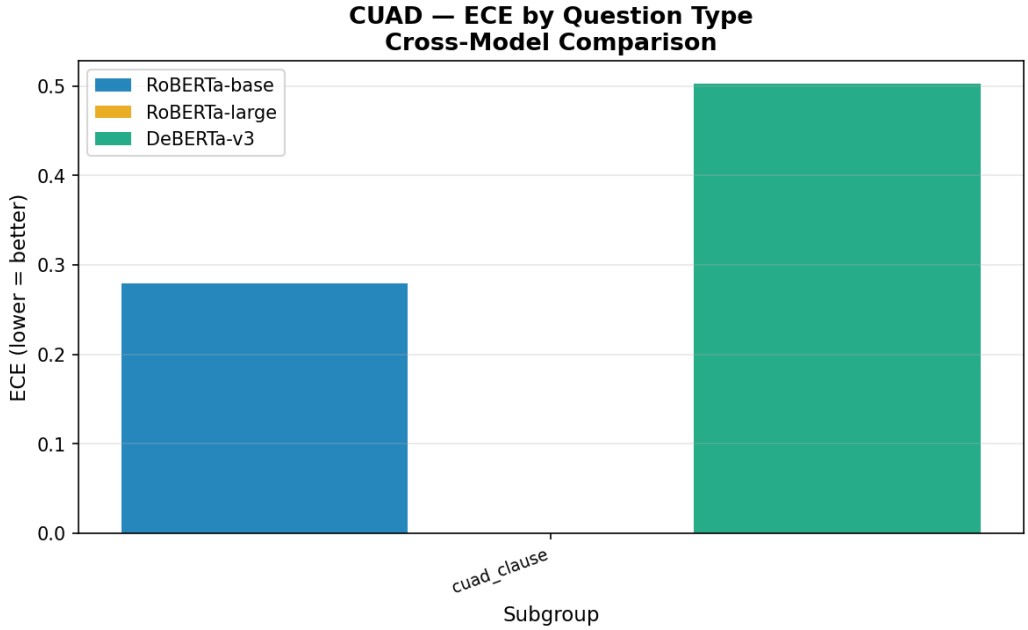


Figure 14. CUAD ECE by question type (cuad_clause), cross-model comparison. DeBERTa's CUAD ECE (0.503) is nearly double RoBERTa-base's (0.279), indicating worse calibration under legal domain shift.

8.6 CUAD — Subgroup Calibration by Answer Length

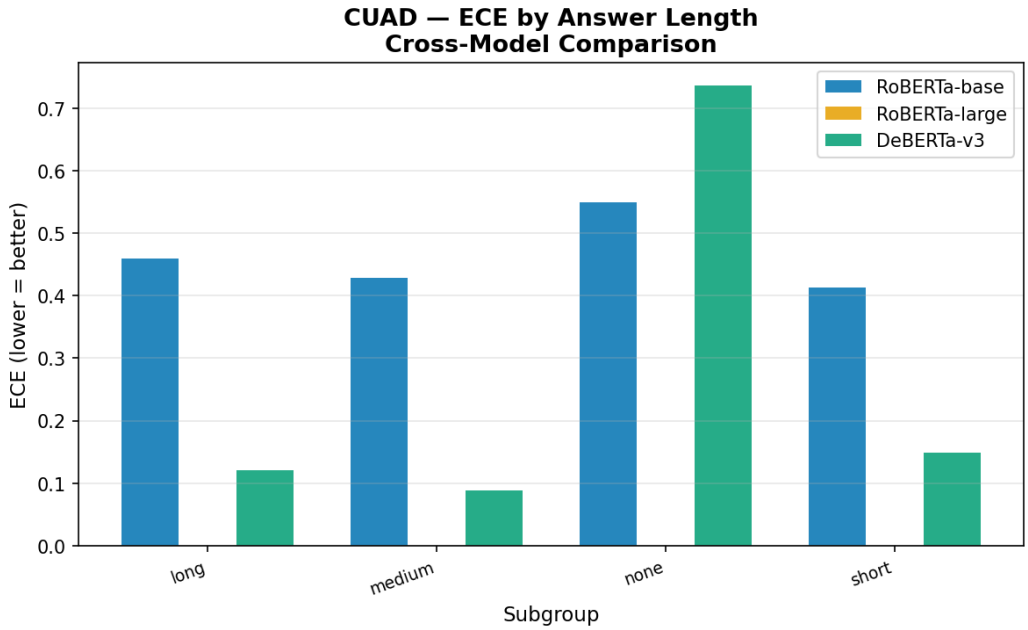


Figure 15. CUAD ECE by answer length, cross-model comparison. DeBERTa's unanswerable ECE (0.737) is far worse than RoBERTa-base's (0.550), while DeBERTa achieves lower ECE on answerable subgroups — reflecting a different failure mode under domain shift.

Subgroup	RoBERTa-base ECE	RoBERTa-base Acc.	DeBERTa ECE	DeBERTa Acc.
Answerable	0.445	**0.6%**	**0.107**	4.3%
Unanswerable	0.550	**99.1%**	**0.737**	84.4%
Long answers	0.460	**0.0%**	**0.121**	**0.0%**
Short answers	0.413	3.5%	**0.149**	**15.7%**

Table 13. CUAD subgroup calibration cross-model. Bold values highlight worst ECE and extreme accuracy values. Both models fail on answerable CUAD examples; DeBERTa answers more but is less consistently right.

8.7 CUAD — Subgroup Calibration by Answerability

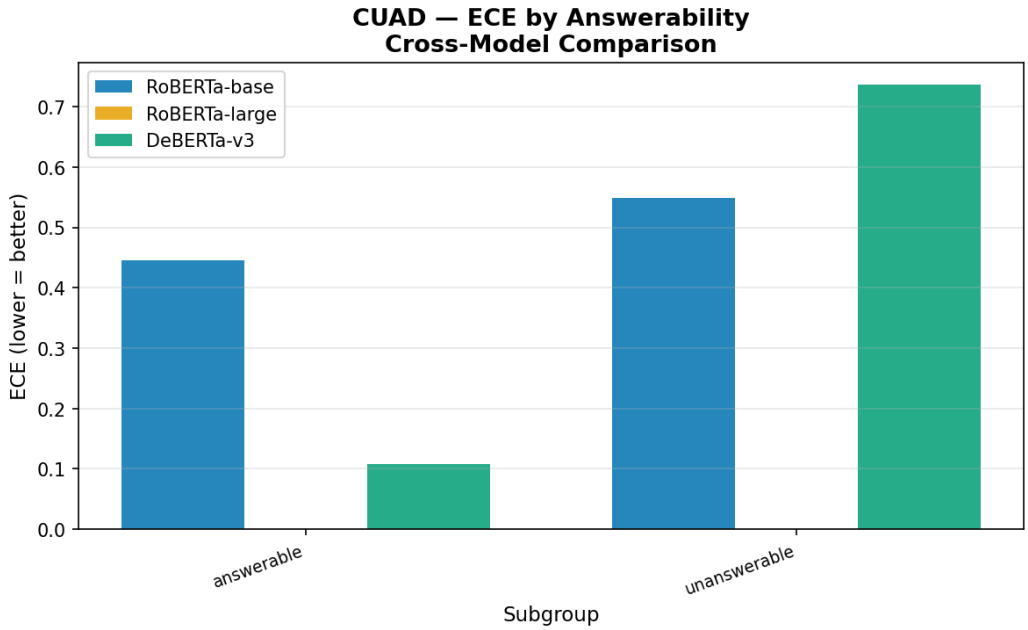


Figure 16. CUAD ECE by answerability, cross-model comparison. RoBERTa-base nearly abstains on everything (99.1% unanswerable accuracy). DeBERTa attempts more answers but is less reliably calibrated on unanswerable examples.

9. What We Have Built

Beyond the research findings, we have built a clean, reproducible codebase on GitHub:

File	Purpose	Status
requirements.txt	All dependencies for the full 8-week project	✅ Done
src/setup_check.py	Environment verification — 16/16 checks passing	✅ Done
src/load_data.py	Loads SQuAD2.0 from HuggingFace + CUAD from local zip	✅ Done
src/run_inference.py	Full inference pipeline saving logits + confidence signals	✅ Done
src/evaluate_baseline.py	F1/EM evaluation with benchmark comparison	✅ Done
src/subgroup_tagger.py	Tags every prediction across 4 subgroup axes	✅ Done
src/calibration_analysis.py	ECE, reliability diagrams, confidence distributions	✅ Done
src/cross_model_analysis.py	Cross-model ECE comparison and figures (NEW)	✅ Done

Table 14. Codebase files and status. All scripts are complete and producing publication-ready outputs.

10. Forward Plan - Roadmap

Immediate — RoBERTa-large Inference (In Progress)

RoBERTa-large inference is currently running on the UMD Zaratan HPC cluster using 16 CPU cores. Once complete, all cross-model figures and tables will be updated to include the third model. This is expected to complete imminently.

Week 3 Completion — Statistical Validation

Bootstrap confidence intervals on all ECE differences. Compute AUC-RC per subgroup for all three models. Without statistical testing reviewers will not accept subgroup differences as significant findings.

Week 4 — LLM Calibration Comparison

This is the most novel and ambitious component. Run a small open-source LLM (Llama-3.2-3B or Mistral-7B) on the same SQuAD and CUAD examples. Extract confidence signals from:

- Token logprobs — the log probability of the generated answer tokens
- Verbalized uncertainty — prompt the model to say how confident it is (0–100%)
- Consistency sampling — run the same question multiple times, measure answer variance

Then apply the same calibration analysis. The research question: do LLMs, despite higher F1, produce better-calibrated uncertainty estimates than fine-tuned extractive models? This is genuinely open — there is no existing systematic answer in the literature.

Weeks 5–6 — Cross-Model Trust Module

Train the trust module (logistic regression or gradient boosted trees) on confidence signals from all models. Key questions: do features that matter for RoBERTa also matter for DeBERTa? Can a single trust module generalise across model families? Per-subgroup threshold tuning. Evaluate against temperature scaling and Platt scaling baselines.

Week 7 — Benchmark Package

Config-driven evaluation suite: input any HuggingFace QA model, get the full calibration audit — ECE, reliability diagrams, risk-coverage curves, subgroup breakdown. Model-agnostic design. Public GitHub release with documentation.

Week 8 — Paper Writing

9. Summary of Key Findings So Far

- SQuAD F1 = 82.50 — pipeline validated against reported benchmarks ✓
- Global ECE on SQuAD = 0.1613 — model is miscalibrated even on its training domain
- Yes/No questions (ECE=0.2951) and Why questions (ECE=0.2359) are the most miscalibrated subgroups — nearly 2x worse than How questions (ECE=0.1499)
- Long-answer questions (ECE=0.2274) are significantly worse calibrated than short answers (ECE=0.1462)
- Under domain shift (CUAD), miscalibration becomes catastrophic: answerable ECE = 0.9750, near the theoretical maximum
- The model's confidence distribution is nearly identical for answerable and unanswerable CUAD examples — confidence is useless as an abstention signal out-of-domain
- Miscalibration is structured and predictable — this is the central claim of the paper

10. Key References

The following papers provide the theoretical and empirical foundations for this project. Each is cited in the relevant section of this report.

[1] Guo et al. (2017) — Calibration foundation. On Calibration of Modern Neural Networks. ICML 2017. arXiv:1706.04599. Introduced ECE as a calibration metric, reliability diagrams, and temperature scaling. Directly underpins our calibration methodology and our temperature scaling baseline in Phase 2.

[2] Rajpurkar, Jia & Liang (2018) — SQuAD 2.0. Know What You Don't Know: Unanswerable Questions for SQuAD. ACL 2018. arXiv:1806.03822. Introduced SQuAD 2.0 with adversarially crafted unanswerable questions. Our primary dataset and the source of our benchmark F1 numbers.

[3] Hendrycks et al. (2021) — CUAD. CUAD: An Expert-Annotated NLP Dataset for Legal Contract Review. NeurIPS 2021. arXiv:2103.06268. Our domain shift dataset. 13,000+ expert annotations across 510 legal contracts in SQuAD-compatible format.

[4] Kamath, Jia & Liang (2020) — Selective QA under domain shift. Selective Question Answering under Domain Shift. ACL 2020. arXiv:2006.09462. Showed that softmax-based abstention fails under OOD conditions and proposed a calibrator-based fix — the direct precursor to our trust module.

[5] Varshney et al. (2022) — Selective prediction in NLP. Investigating Selective Prediction Approaches Across Several Tasks in IID, OOD, and Adversarial Settings. Findings of ACL 2022. Formal framework for risk-coverage tradeoffs in NLP — underpins our Week 2 risk-coverage curve analysis.

[6] Xiong et al. (2024) — LLM calibration survey. A Survey of Confidence Estimation and Calibration in Large Language Models. NAACL 2024. arXiv:2311.08298. Comprehensive survey of LLM confidence estimation methods including token logprobs, verbalized uncertainty, and consistency sampling — the three signal types we extract in our LLM comparison phase.